

MUSTER.

the war-room that musters its own responders

A real Kubernetes incident fires → MUSTER musters a minimal shortlist and keeps **only** the specialist that incident needs (a real Contract-Net contest over Band's primitives) → repairs the real cluster with reversible actions, human-gated on anything destructive.

Band coordination layer

Contract-Net (FIPA)

LangGraph · CrewAI · Pydantic AI

Kubernetes RBAC

THE PROBLEM

Incident response is a **fixed panel** with the safety off

- ▶ Most multi-agent demos show a **fixed** roster of agents talking – the same panel convenes for every problem, whether it fits or not.
- ▶ An unscoped operator (human or agent) under pressure touches things it shouldn't – off-limits namespaces, irreversible scale-downs – and the blast radius is the whole cluster.
- ▶ "Human-in-the-loop" is usually a label on a slide, not a key the human actually holds at the moment of a destructive action.

Enterprise incident coordination needs the **right responders only**, real repair, and a hard safety boundary – verifiable, not asserted.

THE DISPLACEMENT

No roster. The **incident** decides who is mustered.

Field's modal solution

A fixed 3–5 agent governance/review panel that always convenes and chats.
Coordination is decorative; the roster is static.

MUSTER

Reads the fault signature, scores responder **fit** (Jaccard of symptom vs capability tags), and musters **only** the matching responders via Band – a `workload-only` signature musters just the `WorkloadResponder`.

Selective muster + reversible real infra + a human-held key + a public URL the judge can fire themselves – that is the differentiation, measured not claimed.

HOW IT WORKS

Contract-Net, mapped onto **real Band primitives**



Commander + responders run as **distinct Band identities** – coordination genuinely flows as Band chat events & messages between them, not in-process calls. Responders run on three **real runtimes** (LangGraph / CrewAI / Pydantic AI) on one coordination substrate.

Honest scope: the bid rule is one deterministic `policy.make_bid`; runtimes differ, the decision rule is shared (reasoning lives in the Commander).

SAFETY THAT THE CLUSTER ENFORCES

The boundary is the **Kubernetes API server**, not Python

- ▶ **Scoped by RBAC.** The hardened responder authenticates as a namespaced ServiceAccount (`responder-shop`, Role+RoleBinding in `shop` only).
- ▶ **Off-limits is API-Forbidden.** Any read/write to the `billing` namespace returns **403 Forbidden** from the API server – regardless of what the agent code attempts.
- ▶ **Reversible only.** Every remediation is a `kubectl` action with a known inverse (rollout undo / scale / image revert). No simulation.
- ▶ **Human holds the key.** Destructive kinds (e.g. `scale --replicas=0`) block on a real human ack – the gate refuses, it doesn't just warn.

Proof on the live cluster: `python -m common.k8s rbac` → **boundary_holds: true** (shop patch ok · billing read 403 · admin control can still read billing).

THE PROOF – MEASURED, REPEATABLE

Same real fault. Naive blast 2 → hardened blast 0.

	naive control (full access, no scope, no gate)	hardened muster (MUSTER)
who acts	one operator, full cluster access	only the mustered responder, scoped to its namespace
destructive op	runs unapproved (scale 0)	blocked, waits for the human key
off-limits billing	touched (deploy/ledger)	untouched
blast radius	2	0
recovered	✗	✓

Re-derive it from scratch in one command: `bash scripts/demo.sh` – brings up the kind cluster, runs the contrast on the same injected fault (exits non-zero unless it holds), prints the RBAC self-test. No Band credentials needed.

WHY IT MATTERS

Coordination you can **audit**, repair you can **trust**

- ▶ **Less manual coordination.** The incident signature recruits the right specialists automatically – no human paging a fixed on-call panel.
- ▶ **Bounded blame radius.** RBAC-scoped responders mean a misfire can't cross into regulated/off-limits systems – a hard requirement for high-stakes enterprise ops.
- ▶ **Trustworthy autonomy.** Reversible actions + a human-held destructive key make agentic remediation deployable where "let the agent run" otherwise isn't.
- ▶ **Transparent by construction.** Every muster, bid, award and tool call streams to a public observatory – the coordination is the evidence.

Fire it yourself.

Pick the workload incident · flip naive ↔ hardened · watch blast
2 → 0 on the real cluster.

Live observatory: [see submission page](#)

Demo video: [see submission page](#)

Repo: MIT · one-command reproducer

Built on Band as a genuine coordination layer – discover · bid · award · handoff – not a thin wrapper.